

LA ROCHE-SUR-YON AP, France

WMO: 073060

Lat: 46.700N

Lon: 1.379W

Elev: 299

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.7	(c) 27.7	(d) 19.2	(e) 14.6	(f) 26.9	(g) 22.9	(h) 17.5	(i) 29.3	(j) 28.5	(k) 50.9	(l) 25.3	(m) 50.6	(n) 7.1	(o) 30	(p) 0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.6	(c) 87.1	(d) 68.5	(e) 83.6	(f) 67.0	(g) 79.9	(h) 65.7	(i) 70.7	(j) 82.7	(k) 68.8	(l) 79.7	(m) 67.2	(n) 76.7	(o) 8.2	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.8	99.8	74.3	65.3	94.4	72.3	63.8	89.7	70.7	34.7	82.6	33.1	79.8	31.8	76.9	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.3	19.3	17.0	DB	20.4	93.9	4.2	3.2	17.3	96.2	14.9	98.0	12.5	99.8	9.4	102.1	
			WB	19.5	73.6	4.1	2.0	16.6	75.0	14.2	76.2	11.9	77.3	8.9	78.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	42.7	43.4	47.4	51.2	57.2	63.5	66.7	66.4	61.8	56.3	48.3	43.6	(6)	(7)
	DBStd	10.52	6.93	6.43	5.58	5.53	5.40	5.77	5.07	5.04	5.25	5.85	6.19	7.10	(7)	(8)
	HDD50	929	234	193	118	48	5	0	0	0	0	16	100	215	(8)	(9)
	HDD65	4245	690	606	547	415	250	96	38	39	125	274	501	664	(9)	(10)
	CDD50	2423	9	8	36	84	227	404	519	507	353	210	50	16	(10)	(11)
	CDD65	260	0	0	0	0	7	50	91	81	28	3	0	0	(11)	(12)
	CDH74	2514	0	0	0	11	98	525	854	764	248	14	0	0	(12)	(13)
	CDH80	772	0	0	0	0	15	156	276	267	56	2	0	0	(13)	(14)
(14) Wind	WSAvg	8.4	9.3	9.6	9.5	8.9	8.3	8.0	7.9	7.4	7.5	7.6	8.3	9.1	(14)	(15)
(15) Precipitation	PrecAvg	33.9	3.9	2.8	2.3	2.6	2.3	1.6	1.9	2.0	2.8	4.0	4.1	3.8	(15)	(16)
	PrecMax	48.7	7.6	6.3	6.0	8.2	4.8	3.9	5.6	4.8	11.0	7.9	10.7	6.8	(16)	(17)
	PrecMin	22.8	0.7	0.2	0.2	0.6	0.4	0.2	0.2	0.2	0.1	0.6	1.0	0.6	(17)	(18)
	PrecStd	6.3	1.9	1.7	1.5	1.6	1.0	0.8	1.1	1.3	2.4	1.8	2.1	1.8	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	60.5	68.8	75.5	82.2	90.5	92.0	93.2	85.9	75.4	63.7	58.2	(19)	(20)
		MCWB	54.9	52.3	55.8	60.7	66.7	70.2	69.8	70.9	66.6	63.9	58.7	55.4	(20)	(21)
	2%	DB	55.3	56.0	63.6	70.3	76.5	84.3	86.7	86.9	80.7	70.5	60.8	56.4	(21)	(22)
		MCWB	53.2	51.0	54.0	59.2	63.8	67.7	67.9	68.9	64.6	62.2	57.8	54.3	(22)	(23)
	5%	DB	53.7	53.6	59.5	65.6	72.3	80.0	82.7	82.4	76.3	67.3	58.9	55.0	(23)	(24)
		MCWB	51.7	50.5	52.2	55.7	61.4	65.9	66.7	67.1	63.5	60.9	56.5	53.1	(24)	(25)
	10%	DB	52.1	52.1	56.2	61.4	68.4	75.8	78.6	77.8	72.4	64.7	57.1	53.5	(25)	(26)
		MCWB	50.3	49.6	51.0	53.7	59.4	63.9	65.5	65.3	61.7	59.8	54.7	51.7	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.9	53.9	57.3	62.9	68.4	72.7	72.3	74.3	69.6	65.7	60.9	56.5	(27)	(28)
		MCDB	56.4	57.0	65.2	72.8	79.2	85.0	86.3	87.6	79.7	70.6	62.0	57.3	(28)	(29)
	2%	WB	53.7	52.5	55.0	59.7	64.9	69.5	69.8	70.9	67.0	63.7	58.5	54.9	(29)	(30)
		MCDB	54.8	54.7	61.0	68.7	74.1	80.5	82.4	82.1	75.7	68.2	60.1	56.0	(30)	(31)
	5%	WB	52.1	51.3	53.5	56.7	62.3	67.2	68.0	68.8	65.0	62.1	56.8	53.5	(31)	(32)
		MCDB	53.3	53.2	57.2	63.7	70.4	77.5	79.4	78.8	72.6	66.1	58.6	54.8	(32)	(33)
	10%	WB	50.5	50.0	52.0	54.5	60.2	65.0	66.5	66.8	63.3	60.6	54.9	51.9	(33)	(34)
		MCDB	51.9	51.9	55.3	60.2	67.1	74.0	76.3	74.5	70.0	63.7	56.9	53.3	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	12.3	15.0	16.7	17.2	18.8	18.6	19.3	18.6	14.2	11.3	10.5	(35)	(36)
		MCDBR	9.4	13.3	20.0	22.9	24.0	25.9	26.5	27.2	24.3	16.6	11.6	9.8	(36)	(37)
	5% WB	MCWBR	8.0	9.8	12.5	13.4	13.1	11.8	11.1	11.1	11.0	9.5	8.8	8.4	(37)	(38)
		MCWBR	8.7	11.6	15.5	20.8	21.7	23.2	22.4	22.5	19.9	13.9	10.2	9.2	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.327	0.342	0.369	0.390	0.393	0.403	0.395	0.384	0.364	0.362	0.337	0.322	(39)	(40)
		taud	2.441	2.394	2.320	2.265	2.305	2.312	2.335	2.381	2.416	2.426	2.436	2.422	(40)	(41)
	Ebn at Noon		241	261	270	274	277	274	274	273	269	250	234	228	(41)	(42)
		Edh at Noon	22	28	35	40	40	40	38	35	31	27	22	20	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		369	666	1033	1486	1751	1964	1886	1669	1319	780	450	323	(43)	(44)
	RadStd		36	68	113	151	136	165	95	129	89	65	44	38	(44)	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page